

Uncertainty-aware haplotype phasing of genetic and epigenetic variation from long reads

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Abstract

Read-based phasers of long reads, which recover haplotypes from a single sample, weigh all observations alike, mostly ignore all but single-nucleotide variants and report phase as a binary state. Here we present LongPhase 2, which co-phases SNVs, indels, structural variants and allele-specific methylation in one graph, weights each observation by base quality and sequence context, and scores every variant’s phasing uncertainty. A graph neural network then unphases, rather than flips, the variants it predicts to be misphased. On HG002 nanopore data (10–60×), against a telomere-to-telomere truth set, LongPhase 2 makes 2.8- to 4.8-fold fewer switch errors than WhatsHap and HapCUT2 (1.1- to 1.7-fold under GIAB v4.2.1) and runs six- to eightfold faster at 60×; co-phasing lengthens haplotype blocks by 40%. On PacBio HiFi data, without retraining the network, LongPhase 2 makes 2.5- to 3.5-fold fewer switch errors than WhatsHap. LongPhase 2 is a drop-in replacement for the LongPhase versions in Clair3, ClairS, Severus and EPI2ME.

Introduction

Haplotype phasing, the assignment of heterozygous variants to their parental chromosomes, is essential for interpreting the diploid human genome. Whether two variants act in *cis* or in *trans* decides compound heterozygosity in recessive disease¹, and allele-specific methylation on a phased haplotype reveals its parent of origin². Long-read sequencing has made read-based phasing practical at clinical and population scale. Individual reads span tens to hundreds of kilobases, link distant heterozygous sites directly and carry native base-modification signals; a single library thus yields haplotype blocks orders of magnitude longer than short reads allow^{3,4}. Read-based phasing also needs nothing beyond the sample’s own reads: no sequencing of the parents, no population reference panel of the kind statistical phasing depends on⁵, and none of the additional Hi-C or Strand-seq libraries otherwise used to phase whole chromosomes⁶.

Read-based phasing is classically formulated as minimum error correction and solved by dynamic programming or graph partitioning, as in WhatsHap⁷, HapCUT2⁸ and Margin⁹. These methods centre on single-nucleotide variants (SNVs), although WhatsHap, by re-aligning each read to both alleles, also co-phases small and large indels with them¹⁰. We previously introduced LongPhase¹¹, which recasts phasing as a weighted allele graph, resolves haplotypes at chromosome scale and co-phases SNVs with structural variants (SVs)^{12,13}. WhatsHap and LongPhase

now sit beneath much of the long-read analysis stack. Clair3 phases with WhatsHap by default and with LongPhase as an option, and the somatic callers ClairS and ClairS-TO use LongPhase by default^{14–17}; the EPI2ME wf-human-variation workflow phases every nanopore genome with LongPhase¹⁸; and the SV callers Sniffles2 and Severus take as input the haplotagged alignments that either phaser writes^{12,19}. An error made by the phaser is therefore inherited, without any warning, by every analysis downstream.

Deployment at this scale has exposed three limitations of current phasers. The first is that 5-methylcytosine (5mC), which long reads report alongside the sequence, plays no part in deciding the phase. Where the two haplotypes differ in methylation, as at imprinted loci and at sites of sequence-dependent allele-specific methylation, a read’s CpG states tie it to one haplotype even where it spans no heterozygous variant, which is where sequence-based blocks end^{20,21}. Yet methylation-aware tools use that signal only once a sequence phase exists. NanoMethPhase assigns methylation to haplotypes that WhatsHap has already resolved²⁰; MethPhaser and HapBridge learn haplotype-specific CpGs from the reads an SNV phaser has tagged, then join adjacent blocks or flag suspected switches^{21,22}; and LongHap, which co-phases SNVs, indels and SVs, adds methylation only to bridge the gaps its sequence phasing leaves²³. Because the informative CpGs are defined by the preceding sequence phase, its errors propagate into the methylation evidence, and methylation can revise a phase only after it is built. Allele-specific methylation is moreover confined to a small fraction of CpGs, varies between cells and tissues, and is read as a per-molecule probability rather than a base^{20,21}; such sparse, noisy evidence is best weighed together with the sequence alleles on the same reads.

The second limitation is that read evidence is weighted uniformly even though the errors that break haplotypes are not uniformly distributed. Base-calling errors concentrate in homopolymers and tandem repeats, reads misplaced in segmental duplications create spurious linkages, and heterozygous calls in copy-number-altered regions mislead the graph^{24,25}. What follows is the switch error, a flip of haplotype assignment within a block that inverts every inference downstream, from compound-heterozygosity calls to haplotype-aware somatic detection^{16,19}.

The third limitation is that phase is mainly reported as a binary state. Statistical phasers have long attached a posterior probability to every phased genotype⁵, and HapCUT2 prunes variants whose posterior falls below a threshold⁸. That posterior, however, comes from the model that made the phase and assumes correctly placed reads; it cannot flag errors from mis-mapped reads or collapsed paralogues. WhatsHap and Margin report no confidence, and LongPhase reports only simple read support, the number of a read’s alleles on each haplotype and a Phred-scaled tagging quality derived from them, not the reliability of each variant’s phase. Learning-based phasers, built on graph neural networks or deep reinforcement learning^{26,27}, do not close this gap. They learn, partly from simulated reads, to optimize the same fragment-cut or minimum-error-correction objective as heuristic phasers, phase SNVs only and report no confidence, and their authors note that errors from mis-alignment in repetitive regions remain unresolved²⁷.

These limitations have been underestimated, because phasers are benchmarked against Genome in a Bottle (GIAB) truth sets built largely by mapping reads to GRCh38^{28,29}, which can assess only the regions where mapping-based methods already work³⁰. The HG002 v4.2.1 release

omits both sex chromosomes and 12% of the autosomes, mostly satellite arrays and segmental duplications^{29,30}, where mis-mapped reads and collapsed paralogues create false linkages^{24,25}. Its phase is also partial and partly circular: it was recorded only for the 74% of heterozygous variants on which trio-based phasing agreed with Strand-seq and HiFi phasing by WhatsHap²⁹, leaving the variants hardest for a read-based phaser the least likely to be scored. Extending benchmarks into difficult sequence has changed conclusions before. GIAB’s growth from 85% to 92% of GRCh38 exposed eight times more false negatives in a short-read call set²⁹, and the complete CHM13 reference removed tens of thousands of spurious variant calls per sample³¹. The complete diploid T2T-HG002 assembly now lifts this ceiling for phasing³⁰. It resolves both haplotypes of every chromosome, adds 701 Mb of autosomal sequence absent from v4.2.1 and phases every heterozygous site. Projected onto GRCh38³², it provides a benchmark on which phasers can be compared where they are most likely to fail.

Here we present LongPhase 2, which addresses each of these limitations. It places 5mC sites in the same weighted phasing graph as SNVs, small indels and SVs, letting methylation inform the phase while it is being built. It weights every read observation by base quality and sequence context, giving less weight to homopolymers, tandem repeats and copy-number-altered regions. It scores the phasing uncertainty of every variant, and an optional graph neural network unphases the variants it predicts to be misphased, leaving their genotypes unchanged. We evaluate it against both GIAB v4.2.1 and the T2T-HG002-derived benchmark, and it runs as a drop-in replacement for LongPhase in existing pipelines.

Results

LongPhase 2 co-phases SNVs, indels, SVs and methylation in one graph and withholds phase it predicts to be wrong

LongPhase 2 resolves haplotypes in three stages (Fig. 1). All heterozygous evidence carried by the same long reads (SNVs, small indels, SVs and allele-specific 5mC sites called by the built-in `modcall` module) enters a single phasing graph per chromosome, in which each observation is weighted by base quality and by the reliability of its sequence context, down-weighting homopolymers, tandem repeats and copy-number-altered regions (Fig. 1a,b). Each phased variant receives a phasing-uncertainty score, and a graph neural network inspects a local window around every variant whose uncertainty is high (Fig. 1c). Variants it predicts to be misphased are unphased rather than flipped, a step we call GNN correction, and a haplotype block that becomes disconnected is split; genotypes are preserved and no new phase is introduced. The outputs are phased VCFs for every variant class and haplotagged alignments (Fig. 1d).

We evaluated LongPhase 2 on HG002 nanopore R10.4.1 data down-sampled to 10–60× and on HG002 PacBio HiFi data at 10–50× (Methods). Values at 10–20× are means of ten down-sampling replicates; values at 30–60×, all HiFi values and the analyses of unphased variants come from one replicate. Phasing was scored against the GRCh38-based GIAB v4.2.1 benchmark and against the benchmark derived from the complete diploid telomere-to-telomere (T2T) assembly of the same individual (v5.0q), which extends evaluation into segmental duplications, satellites and other regions that v4.2.1 excludes. Unless stated otherwise, results refer to v5.0q and to

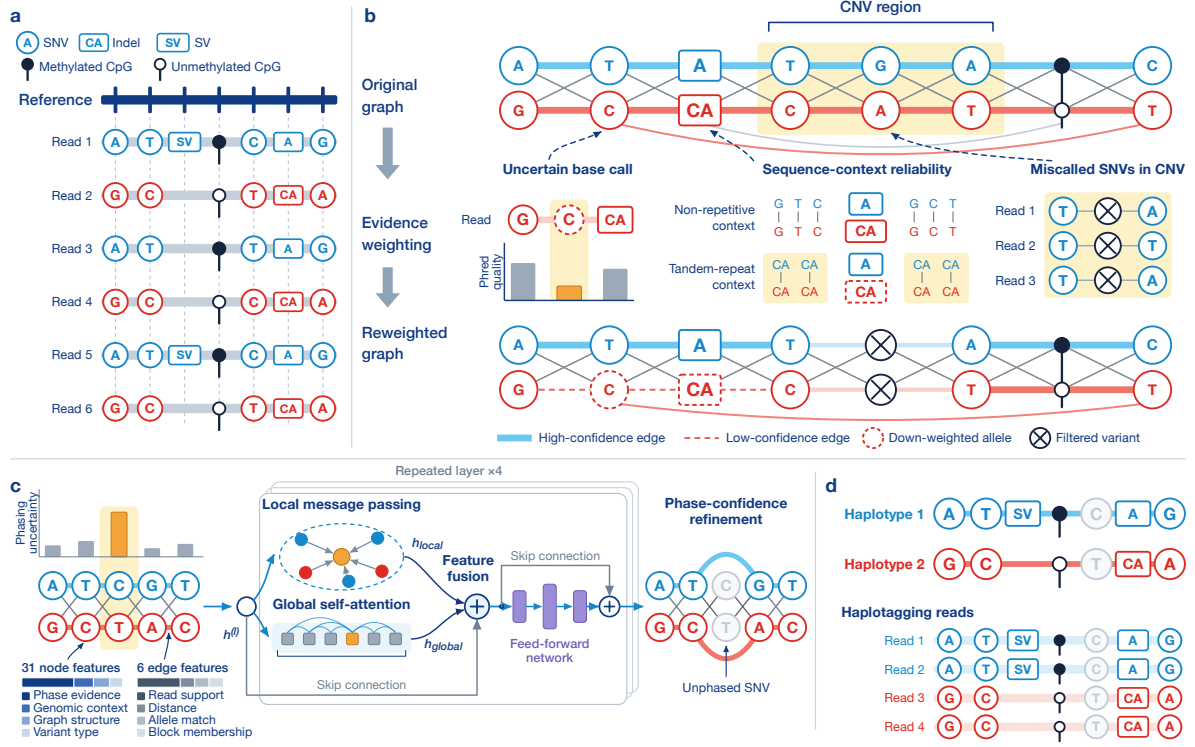


Figure 1: **Overview of LongPhase 2.** **a**, SNVs, small indels, SVs and 5mC sites carried by the same long reads enter one phasing problem; alleles are coloured by read haplotype. **b**, Weighted phasing graph. Read-supported edges link allele nodes, and each observation is weighted by base quality and sequence context; the inset shows an indel down-weighted inside a tandem repeat. **c**, Each variant receives a phasing-uncertainty score (Methods), and a graph neural network (GATv2 and Transformer layers; Supplementary Fig. 6) classifies the variants around the most uncertain ones. Variants predicted to be misphased are unphased (grey), not flipped; we call this step GNN correction. **d**, Output: phased haplotypes of all four variant classes, with the misphased variant left unphased (faded), and haplotagging reads.

SNVs, the only class that every compared tool phases, and LongPhase 2 denotes the complete method, GNN correction included. We first compare LongPhase 2 with existing phasers and then analyse the contribution of its internal modules. Per-run values are in Supplementary Data 1 and Supplementary Tables 7–12.

Comparison with existing phasers

LongPhase 2 makes 2.8- to 4.8-fold fewer SNV switch errors than WhatsHap and HapCUT2. Phasing SNVs only, from identical PEPPER-Margin-DeepVariant calls, LongPhase 2 had a switch error rate of 0.082% at 10× and 0.023% at 60×, against 0.233% and 0.081% for WhatsHap 2.8 and 0.266% and 0.104% for HapCUT2 1.3.4 (replicate s.d. at 10× ≤ 0.006 percentage points), 2.8–3.7-fold below WhatsHap and 3.3–4.8-fold below HapCUT2 across 10–60× (Fig. 2b,e and Supplementary Table 8)[Required validation, not yet run. (i) Genome-wide rates include the 17 chromosomes from which training windows were drawn, on the same down-sampling replicates. Report metrics with and without GNN correction on the held-out chr17, chr21 and chr22 (switch errors, assessed pairs, retained variants, N50; longphase compare -regions) as the primary validation, and state whether these chromosomes influenced the graph constants. (ii) Matched-retention comparison: withhold the same number of variants by the network, by an entropy threshold, by a support-count threshold, at random and by HapCUT2’s per-variant confidence, and plot switch errors against retained variants and N50, separating unphasing from block splitting. (iii) Paired uncertainty (replicate-wise differences) for the fold changes at 10–20×.]. The Hamming distance ranked the tools in the same order (Fig. 2c). WhatsHap phased 0.7–1.0 percentage points more benchmark heterozygous SNVs (Fig. 2a), and the block N50 of LongPhase 2 was slightly shorter than that of the comparators at 10–20× (0.84 versus 0.95 Mb for WhatsHap at 10×) and slightly longer at 30–60× (2.86 versus 2.72 Mb at 60×; Fig. 2d). Whole-genome phasing took 3 min at 10× and 15 min at 60×, against 26 and 95 min for HapCUT2 and 26 and 115 min for WhatsHap [Runtimes have no exact source: they were read from the earlier Fig. 2e (pptx), which the regenerated Fig. 2 no longer contains, and supplementary.xlsx holds none. Supply the timing table (threads, hardware, wall-clock and peak memory per tool and coverage) as a Supplementary Table, or remove this sentence and the speed claim in the abstract. The repository README times phase of v2.0 at 24 threads as 39 s at 10× and 180 s at 60×, about five times faster than the values quoted here]. Of three further phasers compared at 10–20× (Fig. 2f and Supplementary Fig. 10), Ralph²⁷ and GCphase³³ had 2.5–3.5-fold and 21–26-fold higher switch error rates than LongPhase 2 [Hamming distances near 40% are close to random phasing; confirm that GCphase was run with the parameters its authors recommend for nanopore data before reporting it]. Margin⁹ had the lowest rate at 10× (0.069% versus 0.082%), which coincided with 3.2-fold shorter blocks (N50 0.26 versus 0.84 Mb); from 12× upwards LongPhase 2 made fewer switch errors with 2.1–2.8-fold longer blocks.

The separation between phasers depends on the benchmark. Scored against GIAB v4.2.1, the three tools converged on 1,600–2,200 switch errors from 30× upwards, a ratio of 1.2–1.3 between the highest and lowest rate; scored against v5.0q, the same VCFs differed 4.3–4.8-fold (Fig. 3a,b and Supplementary Table 9). HapCUT2 made more switch errors under

v5.0q at every coverage and WhatsHap at every coverage below 60× (both 1.6-fold more at 10×), whereas LongPhase 2 made fewer (1,518 versus 1,820 at 10×; 496 versus 1,676 at 60×). Hamming distance was nearly unchanged between benchmarks (Fig. 3c,d), consistent with the additional v5.0q switch errors being isolated switches; because error locations have not been intersected, the contributions of regional coverage, variant representation and phase discordance remain unresolved[Score the regions and alleles common to both benchmarks separately from the benchmark-specific ones, and intersect switch-error locations across tools and benchmarks].

With indels, LongPhase 2 phases SNVs 3.0- to 3.7-fold more accurately than WhatsHap at similar contiguity. WhatsHap phases indels by default, which lengthened its blocks (N50 2.72 to 4.20 Mb at 60×) but raised its SNV switch error rate by 17–20% (Fig. 4a,b). Phasing the same indels, LongPhase 2 made 3.0–3.7-fold fewer SNV switch errors than WhatsHap, whereas WhatsHap phased 2.4–4.5 percentage points more benchmark heterozygous indels (Fig. 4c) and built 14–28% longer blocks. Adding SVs and 5mC sites, which WhatsHap does not phase, brought the block N50 of LongPhase 2 to within 2–19% of that of WhatsHap with indels (4.10 versus 4.20 Mb at 60×) at 3.0–3.7-fold fewer switch errors (Fig. 4b,d and Supplementary Table 10). MethPhaser²¹, which adds methylation evidence to existing SNV blocks, left the block N50 and switch errors of the LongPhase 2 SNV phasing it was given unchanged, whereas joint SNV and 5mC phasing lengthened blocks by 8% (Supplementary Fig. 11)[The MethPhaser values in MethPhaserCompare.jsx are identical, to every printed digit and at all six coverages, to the SNV-only LongPhase 2 run without GNN correction given to MethPhaser as input (Supplementary Data 1, replicate 1). Confirm that the MethPhaser output VCF was scored and that MethPhaser ran to completion (it is expected to merge blocks); if it did not, re-run it before this comparison is reported].

LongPhase 2 makes 2.5- to 3.5-fold fewer switch errors than WhatsHap on PacBio HiFi data. On SNV-only phasing of HG002 PacBio HiFi data, with the network trained on nanopore data alone, LongPhase 2 had a 2.5–3.5-fold lower switch error rate than WhatsHap (0.052% versus 0.128% at 10×; 0.042% versus 0.125% at 50×), whose rate did not decrease with coverage, and a lower Hamming distance at every coverage (Fig. 5b,c,e and Supplementary Table 11). Block N50 was within 3% of that of WhatsHap, and WhatsHap phased 0.4–0.7 percentage points more SNVs (Fig. 5a,d)[The HiFi call sets are nearly identical at every coverage (2,367,609–2,367,619 heterozygous SNVs from 10 to 50×; Supplementary Table 7), which suggests that variants were called once and the same calls phased at every down-sampled coverage, unlike the nanopore runs. Confirm, and state the HiFi data source, caller and procedure in Methods; if one call set was used, the phased fraction at 10× (87%, against 78% on nanopore) is not comparable between platforms]. HiFi blocks were 4.6 times shorter than nanopore blocks at 50× (0.55 versus 2.51 Mb), as expected from the shorter reads.

LongPhase 2 and WhatsHap leave different SNVs unphased. LongPhase 2 left more heterozygous SNV calls unphased than WhatsHap (210,683 versus 142,313 at 60×), and the two sets overlapped only partly: 88,692 were left unphased by both, 121,991 by LongPhase 2 alone and 53,621 by WhatsHap alone (Fig. 6a). Each tool’s set was enriched in genomic intervals

containing switch disagreements between the other tool and the truth set, increasingly with coverage: 4.3- to 9.6-fold (10 to 60 $\times$) for SNVs left unphased by LongPhase 2 alone and 2.0- to 9.5-fold for those left unphased by WhatsHap alone (Fig. 6b). This measures regional co-location, not site-level correctness. The two sets also differed in depth (Fig. 6c and Supplementary Fig. 12): at 60 $\times$, SNVs left unphased by WhatsHap alone had a median depth of 18 $\times$, whereas 15.6% of those left unphased by LongPhase 2 alone lay at the depth cap of the variant caller, a pile-up typical of reads mismapped from paralogous sequence[the median depths and the zero-entropy share among WhatsHap-only sites (57.6%) are quoted from JHL’s commit message of 2026-09-27 and are not in any committed table; add the per-set summary to Supplementary Data 1].

Contribution of the internal modules

GNN correction removes a third of the remaining switch errors at a small cost in contiguity. Without GNN correction, the graph algorithm alone had a switch error rate of 0.121% at 10 $\times$ and 0.034% at 60 $\times$, already 1.9–2.5-fold below WhatsHap and 2.2–3.3-fold below HapCUT2 (Supplementary Fig. 13 and Supplementary Table 8). GNN correction lowered it by a further 31–33% at every coverage, at a cost of 0.1–0.3 percentage points of phased SNVs and of 5–9% of block N50 at 10–20 $\times$ and 2–5% above. On HiFi data it removed 19–28% of the switch errors at a cost of 0.1 percentage points of phased SNVs and 1–2% of block N50 (Supplementary Fig. 13e,f).

Methylation lengthens blocks at a similar error rate, and indels trade accuracy for contiguity. We phased seven combinations of variant classes, each with and without GNN correction (Supplementary Figs. 14 and 15 and Supplementary Table 10). Without GNN correction, adding 5mC lengthened the 60 $\times$ block N50 by 8%, SVs by 1%, indels by 49% and all three by 75% (2.91 to 5.10 Mb). Only indels raised the SNV switch error rate, from 0.034% to 0.043% at 60 $\times$ and from 0.121% to 0.155% at 10 $\times$ [For configurations that phase indels, compare computes the Hamming distance over all common heterozygous variants, indels included (CompareProcess.cpp:579), whereas switch errors are SNV-only. Every Hamming value of an indel-containing configuration in this section, in Figs. 3c,d and 4d and in Supplementary Tables 9 and 10, therefore mixes SNV and indel phasing. Re-score these runs with SNV-only Hamming distance and report indel phasing accuracy separately]. GNN correction lowered the switch error rate of every configuration by 30–42% and reduced the increase attributable to indels by 41–67%; it also split blocks that unphased indels had bridged, shortening indel-containing configurations by 12–22%. After GNN correction, SNV+5mC phasing had a switch error rate similar to that of SNV-only phasing (497 versus 496 switch errors at 60 $\times$) with 8% longer blocks, and four-class phasing reached 0.027% with 43% longer blocks (4.10 versus 2.86 Mb). LongPhase 2 phased 30.8–45.5% of benchmark heterozygous indels, more than 99% of the 5mC sites called by modcall and 68–70% of the heterozygous SVs called by Sniffles2[The SV and 5mC phased fractions were read from the earlier Fig. 4c,d (pptx) and have no exact source: supplementary.xlsx lists only the called SV and 5mC totals (Supplementary Table 7). Add the phased SV and 5mC counts to Supplementary Data 1 or drop these two clauses].

SNVs withheld by GNN correction are enriched outside assembly-assessable regions.

GNN correction unphased 59,000–64,000 SNVs per genome. Classified against the v5.0q benchmark VCF, 81% (10×) to 93% (60×) were absent from it, 3–5% were homozygous and 14% to 4.0% (8,673 and 2,411) were heterozygous (Supplementary Fig. 16a)[The classification counts benchmark VCF records outside the benchmark BED, whereas compare scores only inside it: compare puts the loss of phased benchmark SNVs at 60× at 2,055 (2,193,366 before and 2,191,311 after GNN correction, Supplementary Data 1), and the assembly analysis below finds only 549 unphased SNVs inside the benchmark BED, of which 413 are benchmark variants. The two analyses also count different totals at 60× (60,267 here, 56,076 in the assembly analysis of the same replicate). State the matching rule and SNV filter of each and reconcile the totals]. Because v5.0q is derived from the T2T-HG002 assembly, absence means either that the assembly carries no variant or that it does not align to GRCh38 at the site. Of 56,076 SNVs unphased at 60×, 11.8% lay where both assembled haplotypes align one-to-one to GRCh38, against 92.6% of a random sample of SNVs phased before GNN correction (odds ratio 94; Supplementary Fig. 16b and Supplementary Table 12). The difference held on every autosome (Supplementary Fig. 16c), and the unaligned sites clustered in pericentromeric satellites, most densely at chr11:51–54 Mb, chr12:35–36 Mb and chr18:15–20 Mb. Of the 6,601 unphased SNVs inside the alignment, 3,421 (52%) had no assembly variant record at the same coordinate[v5.0q is derived from the same assembly; this analysis extends coverage beyond the benchmark BED; it is not independent confirmation. The dipcall BED is the raw alignment and does not exclude the assembly errors that the benchmark curation removes. Rebuild this analysis before submission from one site-level table: autosomes only (chrX and chrY are currently included); a control drawn from SNVs that stayed phased, excluding the removed ones (the current background is sampled from all SNVs phased before GNN correction, run_t2t_validation.sh:75); and an allele-aware comparison with the assembly VCF that handles overlapping complex variants (the current test matches the start coordinate only, line 104). Derive every panel of Fig. 6 from that table]. Removed benchmark heterozygotes often had weak phasing evidence, 43% (10×) to 22% (60×) with a phasing entropy of exactly 0, whereas clustering within 50 bp or low genotype quality marked 46–49% of benchmark-absent calls (Supplementary Fig. 17)[these prevalences were read from the earlier Fig. 6b,c (pptx), now Supplementary Fig. 13; replace with exact values and regenerate the figure].

Discussion

LongPhase 2 rests on two ideas: that every class of heterozygous evidence a read carries belongs in one haplotype model, provided that each observation is weighted by how far it can be trusted, and that a phaser should withhold phase where its own output is unreliable. The first extends observations made for methylation^{2,20,21} and for SVs^{11,34} that SNVs alone leave blocks stranded where heterozygosity is low; unlike LongHap, which adds methylation after sequence phasing to bridge the blocks it leaves²³, LongPhase 2 weighs methylation and sequence evidence together as the phase is built. The evidence classes did not contribute alike. Methylation lengthened blocks at a similar switch error rate, whereas indels contributed most of the additional contiguity and most of the additional error, as they did for WhatsHap; this is consistent with difficulties

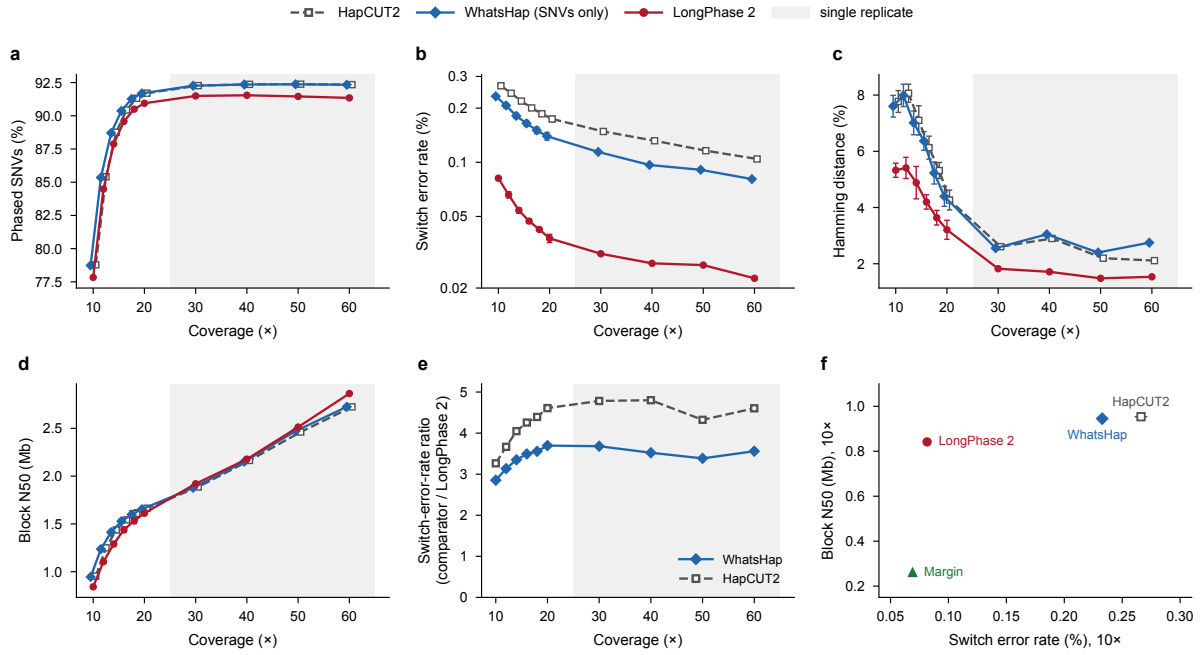


Figure 2: SNV phasing compared with WhatsHap, HapCUT2 and Margin. HG002 nanopore R10.4.1 reads down-sampled to 10–60 $\times$; SNV-only phasing of identical variant calls, scored against the T2T-HG002-derived v5.0q benchmark. Points at 10–20 $\times$ are means of ten down-sampling replicates with s.d.; shaded, single replicate. WhatsHap and HapCUT2 points are offset by $\pm 0.5\times$ for visibility; their values nearly coincide in **a** and **d**. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate (log scale). **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Switch error rate of each comparator divided by that of LongPhase 2. **f**, Switch error rate against block N50 at 10 $\times$; Ralph and GCphase in Supplementary Fig. 10.

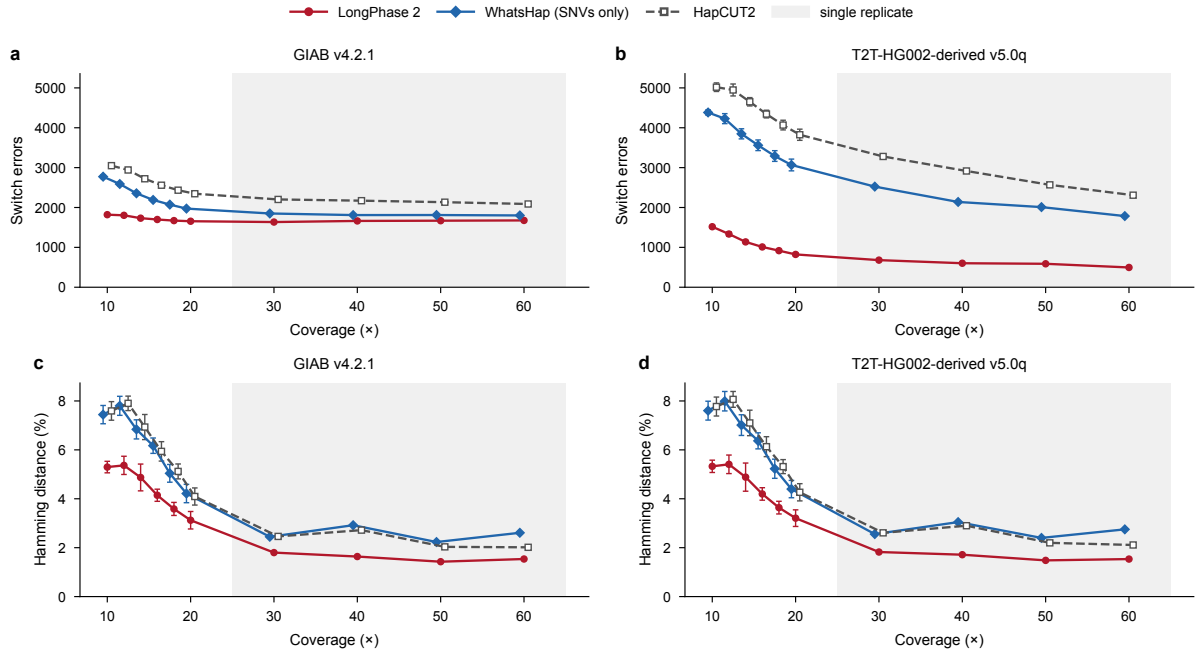


Figure 3: **The same phased VCFs scored against GIAB v4.2.1 and the T2T-HG002-derived v5.0q benchmark.** SNV-only phasing of HG002 nanopore R10.4.1 data; replicates, shading and offsets as in Fig. 2. **a,b**, Genome-wide switch errors under v4.2.1 (**a**) and v5.0q (**b**), on a shared axis. **c,d**, Hamming distance under v4.2.1 (**c**) and v5.0q (**d**).

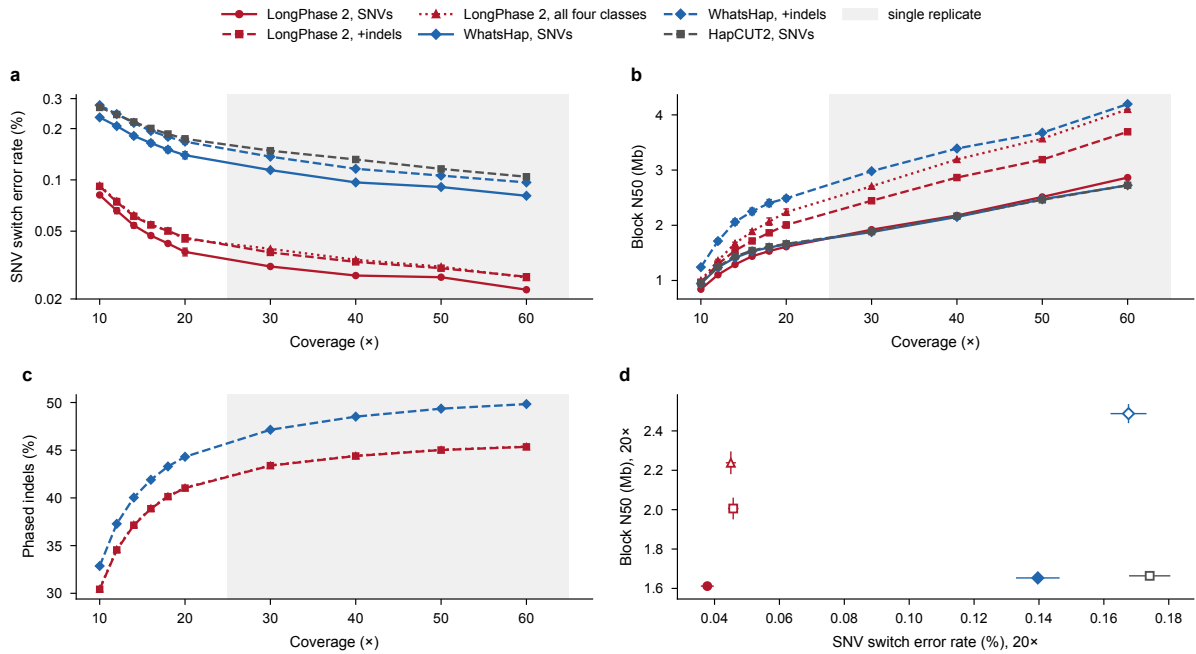


Figure 4: **Co-phasing compared with WhatsHap.** HG002 nanopore R10.4.1 data; LongPhase 2 phasing SNVs only, SNVs and indels, or all four classes (SNVs, indels, SVs and 5mC), against WhatsHap phasing SNVs only or SNVs and indels; HapCUT2 for reference. Accuracy is scored on SNVs against v5.0q; replicates and shading as in Fig. 2. **a**, SNV switch error rate (log scale). **b**, Phase-block N50. **c**, Fraction of benchmark heterozygous indels phased. **d**, SNV switch error rate against block N50 at 20x (means of ten replicates with s.d.).

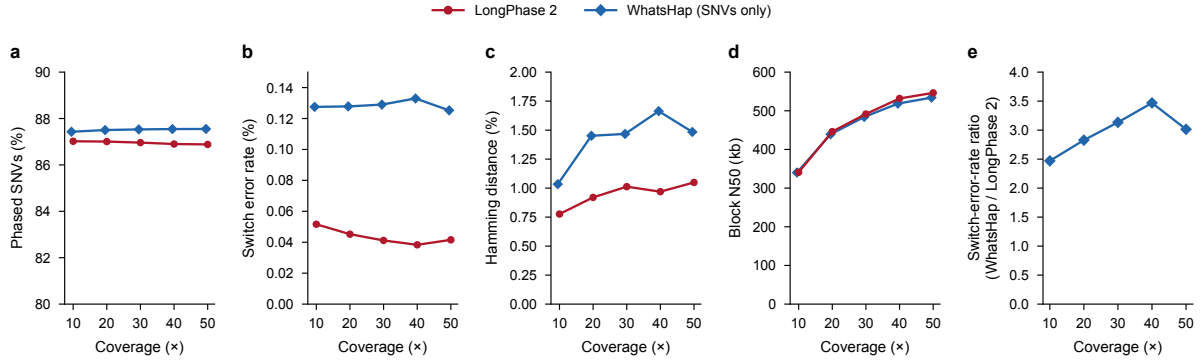


Figure 5: **Phasing of PacBio HiFi data compared with WhatsHap.** HG002 PacBio HiFi (Revio) reads down-sampled to 10–50 $\times$, one replicate per coverage[*state whether one call set was used for every coverage*]; SNV-only phasing scored against v5.0q; LongPhase 2 uses the network trained on nanopore data. Panels as in Fig. 2a–e. **a**, Fraction of benchmark heterozygous SNVs phased. **b**, Switch error rate. **c**, Block-wise Hamming distance. **d**, Phase-block N50. **e**, Switch error rate of WhatsHap divided by that of LongPhase 2.

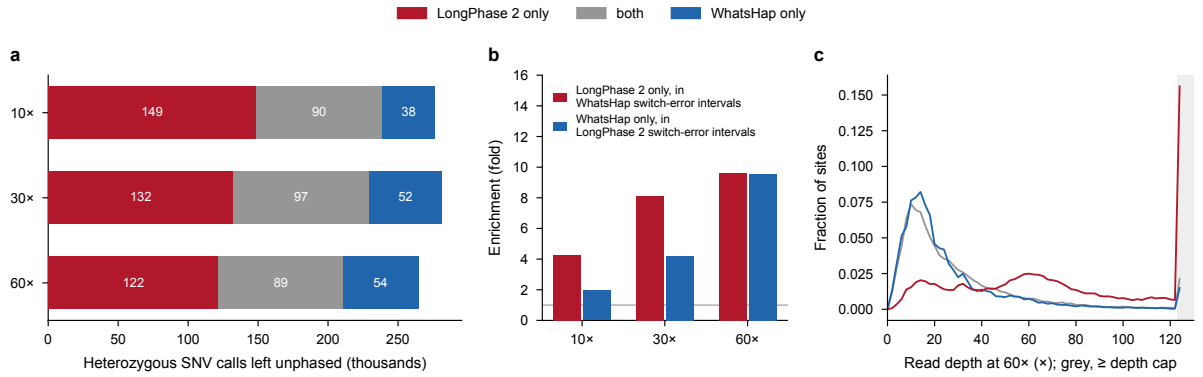


Figure 6: **SNVs left unphased by LongPhase 2 and by WhatsHap.** HG002 nanopore R10.4.1 data, SNV-only phasing, replicate 1; heterozygous SNV calls present in both output VCFs. **a**, Calls left unphased by LongPhase 2 only, by both tools or by WhatsHap only. **b**, Enrichment of each tool-specific set in genomic intervals containing switch disagreements between the other tool and the truth set, relative to that tool's genome-wide rate; a measure of regional co-location, not of site-level correctness. **c**, Read-depth distribution of the three sets at 60 $\times$ (2 $\times$ bins); the grey bin collects sites at or above the depth cap of the variant caller.

introduced by indel evidence, although limitations that both tools share in representing or modelling indels remain possible. Joint phasing is therefore a choice between contiguity and accuracy, which GNN correction narrows and more accurate indel calling^{15,35} would narrow further.

The second idea replaces a binary phased-or-unphased state with a graded one, but only partly. LongPhase 2 writes the phasing entropy and weighted vote totals of every phased variant, in the spirit of the posteriors of HapCUT2⁸ and statistical phasers⁵, yet many removed benchmark heterozygotes had an entropy of exactly 0, unanimous but thin support that a high-entropy threshold alone would not flag. The network reads the evidence across a window rather than the likelihood of the phasing model, as HapCUT2’s pruning does, but whether it withholds better than simpler rules (entropy or support-count thresholds, random removal or HapCUT2’s confidence) at matched numbers of withheld variants has not been tested, and its phase-error score is neither calibrated nor reported. Because GNN correction only withholds phase, part of its gain follows by construction; Margin’s lower switch error rate at 10×, which coincided with three-fold shorter blocks, shows why error must be read alongside contiguity. What GNN correction withheld lay mostly outside the regions that the assembly-derived truth set can assess, where phase cannot currently be verified, and its benefit carried over to HiFi data from the same individual.

Several limitations bound these conclusions. All data come from one individual, HG002; the network was trained on its nanopore data, with labels from the evaluation benchmark and windows from the same down-sampling replicates, and the genome-wide rates include the 17 chromosomes that contributed training windows (Methods). The gains therefore require confirmation on the held-out chromosomes, on an independent individual and benchmark with frozen weights and thresholds, and above 20×[The training-label code matches truth records by position only, without REF/ALT, accepts unphased truth genotypes and compares phase within query phase sets alone (prepare_gnn_data_10.py:343, 440). Audit the training records for allele mismatches, unphased truth and truth-block crossings; if any are found, correct the labels and retrain]. Values at 30–60× and on HiFi come from single replicates, and the HiFi runs appear to share one call set. Benchmark choice changed the magnitude of disagreement and the separation between methods^{29,30}; the contributions of regional coverage, variant representation and phase discordance remain unresolved, and comparisons should report both truth sets and stratify errors by genomic context²⁵. For indels, SVs and 5mC we report how many were phased, not whether they were phased correctly, and corrections to SV and 5mC phase are unsupervised. 5mC co-phasing was assessed only on the autosomes of a male lymphoblastoid cell line²¹, and the comparison with MethPhaser awaits verification. LongPhase 2 assumes a diploid genome with balanced allele fractions.

As a drop-in replacement in pipelines built on LongPhase^{14,16,18,19}, LongPhase 2 adds per-variant phasing evidence without interface changes. The most direct use is in haplotype-aware somatic calling^{16,19}, which accepts inherited phase without qualification: withholding phase at the loci where switch errors concentrate should keep spurious germline phase out of haplotagged tumour reads. Testing that, by varying only the germline phaser upstream of a fixed somatic caller, and providing a calibrated per-variant confidence that downstream tools can read, as they

318 read genotype quality, are the next steps.

319 Methods

320 Overview of LongPhase 2

321 LongPhase 2 takes as input long-read alignments (BAM or CRAM), a reference genome and
322 heterozygous variant calls, and produces phased VCF files for every variant class together with
323 haplotagged alignments (Fig. 1). The pipeline comprises four sub-commands implemented in a
324 single C++ binary linked against HTSLib³⁶: **modcall** identifies allele-specific 5-methylcytosine
325 (5mC) sites from the MM/ML modification tags of the alignments; **phase** co-phases SNVs, small
326 indels, SVs and 5mC sites in one weighted phasing graph per chromosome and emits per-variant
327 phasing diagnostics; the optional **gnn** sub-command applies a graph neural network to the diag-
328 nostics and removes phase assignments that are predicted to be erroneous; and **haplotag** assigns
329 each read to a haplotype using the phased variants. A fifth sub-command, **compare**, implements
330 the evaluation metrics (below). **phase**, **modcall** and **compare** are parallelized over chromo-
331 somes with OpenMP; **gnn** distributes trigger windows over a pool of C++ threads, sharing
332 one immutable copy of the network weights. The graph phasing algorithm extends the design of
333 LongPhase 1¹¹; the sections below describe the current algorithm, with the implementation-level
334 detail of the individual modules in Supplementary Methods 1–6; command-line parameters and
335 their defaults are documented in the repository (Code availability). LongPhase 2 is a germline,
336 diploid phaser.

337 Input variant calls

338 Heterozygous SNVs and small indels were taken from a standard long-read small-variant caller
339 (PEPPER-Margin-DeepVariant⁹ in this study); homozygous and filtered records are carried
340 through to the output unchanged but do not enter the graph. With **-indels**, small indels were
341 phased alongside SNVs; indels with **QUAL** below a user threshold (**-indelQuality**, default 0, i.e.
342 disabled) were excluded and marked in the output **FILTER** column. Each indel was additionally
343 annotated as lying in a tandem repeat, because indel calls in repetitive sequence are the class
344 most often misplaced by the caller and, once misplaced, link the wrong alleles. The dinucleotide
345 immediately 3' of the indel position was read from the reference, and the indel was flagged when
346 that dinucleotide recurred in five consecutive copies. Only dinucleotide units are considered;
347 extending the test to trinucleotide units was evaluated during development, did not reduce
348 switch errors consistently across coverages and was therefore not adopted. Flagged indels were
349 not discarded but retained as phasing evidence, casting votes at one-tenth weight (see below).
350 At that weight they can still bridge SNV-poor regions without dominating a contested decision.
351 SVs were taken from Sniffles2¹² or cuteSV¹³ run with read-name reporting (**-output-rnames**
352 or **-report_readid**); homozygous SVs, SVs whose breakpoint coincides with an SNV, and SVs
353 with duplicated positions were excluded. For each retained SV we recorded the set of supporting
354 read names and the SV length.

Calling allele-specific methylation (`modcall`)

Base-level modification probabilities (Supplementary Fig. 7) were read from the MM/ML tags produced by Dorado³⁷ and carried into the alignments by `minimap2 -y`³⁸; PacBio HiFi reads carry the same tags, but 5mC was not called on the HiFi data used here. For each CpG on each read, the base was labelled methylated if the 5mC probability was ≥ 0.8 (`-modThreshold`) and unmethylated if that probability was ≤ 0.2 (`-unModThreshold`); intermediate values were treated as noise. Only the 5mC probability (modification code `m`) was read (`ModCallParsingBam.cpp:169`). 5hmC calls (code `h`) were ignored. Under a 5mCG_5hmCG basecalling model, a cytosine called as 5hmC was counted as unmethylated whenever its 5mC probability was ≤ 0.2 . Reads on the reverse strand were mapped to the forward-strand CpG coordinate and the two strands of each CpG were genotyped jointly. Each read was thereby assigned to one of three states at the site, and a site was called heterozygous (allele-specifically methylated) when the two states were balanced and the ambiguous fraction small:

$$\frac{\min(M_R, U_R)}{\max(M_R, U_R)} \geq 0.6 \text{ (-heterRatio)}, \quad \frac{A_R}{M_R + U_R + A_R} \leq 0.2 \text{ (-noiseRatio)}, \quad (1)$$

where M_R , U_R and A_R are the numbers of methylated, unmethylated and ambiguous reads; otherwise the site was called homozygous methylated or unmethylated and excluded from phasing. Only CpG dinucleotides were considered; modification calls at non-CpG cytosines were discarded.

A site passing equation (1) is balanced but not necessarily allele-specific, because a site at which methylation varies stochastically between reads satisfies the same inequalities. Candidate sites were therefore required to co-segregate with a neighbouring heterozygous marker. Writing RR , RA , AR and AA for the numbers of reads carrying each of the four combinations of the two states at a pair of sites, a pair was accepted as linked when

$$\frac{\max(RR + AA, RA + AR)}{RR + AA + RA + AR} \geq 0.9 \text{ (-connectConfidence)}, \quad (2)$$

that is, when nearly every read supports one of the two possible pairings rather than a mixture. Equation (2) is the criterion the phasing graph itself applies to a pair of alleles. As a result, the two methylation states enter the graph as reference and alternate alleles do, and `phase` requires no methylation-specific machinery.

Linking was applied in two modes: *SNV-anchored*, used whenever an SNV VCF was supplied, in which a candidate CpG must satisfy equation (2) against an SNV among the next 20 variants (`-connectAdjacent`); and *methylation-only*, used where no informative SNV is in range, in which the same criterion is applied between two candidate CpGs. Candidates that never satisfied it were held back as weak points and admitted only on a later pass against an already accepted CpG. Runs of consecutive CpG coordinates were then merged into a single locus represented by the first position, which consolidates the forward- and reverse-strand calls of a run into one read list and removes the need for `phase` to reason about strand. The resulting VCF records the methylated and unmethylated read names (`MR`, `NR`), the read strand and the per-site depths, and is passed to `phase` through `-mod-file`. The depth bound on the neighbour scan, the weak-

point expansion and the rationale for merging are given in Supplementary Method 1. The values quoted here are the compiled-in defaults.

Extraction of read-level allele observations

For each chromosome, all alignments with mapping quality ≥ 1 (`-mappingQuality`) that were not unmapped, secondary or marked as optical/PCR duplicates were streamed once and, for every variant they overlap, converted into an allele observation with an associated quality. Supplementary alignments were retained as phasing evidence, which is why the overlapping-alignment filter below is required; this differs from the stricter filters applied inside `modcall` and `haplotag`. SNV alleles were read from the aligned base and inherited its Phred base quality; indel alleles were called from the CIGAR string; SV alleles were assigned by scanning a window of 20 CIGAR operations (`-svWindow`) around the SV start for an insertion or deletion whose length is within 10% of the SV length (`-svThreshold`), and reads named by the SV caller were also assigned the ALT allele; and 5mC alleles were taken from the `modcall` read lists. Soft- and hard-clipping operations longer than 5 bp were counted per reference position, separately for clips at the front (left) and back (right) end of an alignment, as evidence of alignment breakpoints for copy-number detection (below). Supplementary Method 2 gives the per-class extraction rules in full.

Three filters were applied before graph construction. (1) *Homopolymer SNV filter*, applied under `-ont` only: when two SNVs both lie in homopolymers of length ≥ 3 and are ≤ 2 bp apart, the downstream SNV was removed, because such pairs are dominated by systematic homopolymer-length errors in nanopore reads. (2) *Overlapping-alignment filter*: when two alignments of the same read overlapped by more than 20% of their combined span (`-overlapThreshold`), the shorter alignment was discarded. (3) *Copy-number-aware filter* (Supplementary Fig. 1e,f). Collapsed paralogues and copy-number-altered segments recruit reads from more than one locus; paralogue-specific differences are then called as heterozygous variants and link alleles that are not on the same chromosome. Such segments are bounded by alignment breakpoints. Two boundary shapes were detected from the per-position front- and back-clip counts by a three-state scanner: a *step* boundary, characteristic of a copy-number transition, declared on a cluster of at least five front-clips at one position, and a *ramp* boundary, characteristic of the progressively nested breakpoints left by breakage–fusion–bridge cycles, declared where front-clips merely exceed back-clips. Both active states carried a running excess of front-over back-clips, extended the candidate interval by a 30 kb look-ahead while that excess was large, and closed on a back-clip threshold scaled to the strength of the breakpoint that opened the interval; candidates longer than 200 kb, or in which the excess fell to zero, were abandoned. Supplementary Method 2 gives the state machine, its five constants and the closing rules in full.

Within each interval, every read was scored by its *mismatch load* R_m , the number of heterozygous sites inside the interval at which the read carries the alternate allele, and every heterozygous variant v in the interval was then scored by

$$\text{MR}(v) = \frac{\bar{R}_m^{\text{ALT}}}{\bar{R}_m^{\text{REF}} + \bar{R}_m^{\text{ALT}}}, \quad (3)$$

where $\bar{R}_m^{\text{ALT}}$ and $\bar{R}_m^{\text{REF}}$ are the mean loads over the reads supporting the alternate and the reference allele of v . Variants with $\text{MR}(v) \geq 0.7$ were removed from all reads; Supplementary Method 2 explains why the statistic separates genuine heterozygous sites from paralogue-specific differences.

Weighted phasing graph

Each heterozygous variant v contributes two nodes (Supplementary Figs. 1–2), v^{ref} and v^{alt} . For every read, the observed alleles were sorted by position and each allele node was connected to the allele nodes of the next $k = 35$ variants on the same read (`-connectAdjacent`); no distance limit is applied to these edges or to the votes cast along them, allowing a single read to link variants more than 300 kb apart. The `-distance` parameter (300 kb) acts instead on the variant sequence: a variant whose next heterozygous variant lies further away than this is skipped during haplotype resolution, casting no votes and receiving no haplotype or entropy, which prevents a block from being carried across a gap in heterozygosity. Each read contributes a weight of 1 to the edge if both of the two observations have Phred quality ≥ 12 (`-baseQuality`) and a weight of 0.1 (`-edgeWeight`) otherwise. Observations without a base quality were assigned a fixed one (`PhasingGraph.cpp`:838–862): 60 for indel and 5mC observations and for SV observations carrying the ALT allele, and 30 for SV observations on reads that neither the CIGAR scan nor the caller’s read list assigned to the ALT allele, which are taken to carry the reference allele. Because both values exceed the default base-quality threshold, every indel, SV and 5mC observation contributes a weight of 1 at default settings; the lower quality of reference-allele SV observations reduces their weight only when `-baseQuality` exceeds 30. For a pair of variants (u, v) this yields four accumulated weights $w_{rr}, w_{ra}, w_{ar}, w_{aa}$, the total support for the two allele pairings $P = w_{rr} + w_{aa}$ (*cis* between reference alleles) and $Q = w_{ra} + w_{ar}$, and their similarity $s = \min(P, Q) / \max(P, Q)$.

Haplotype resolution by multi-neighbour voting

Variants were processed in coordinate order. For the current variant u , each of its k downstream neighbours v received a vote for the pairing with the larger support (P versus Q). A pair was declared uninformative and cast no vote if $s > 0.7$ (`-edgeThreshold`); for SNV–5mC pairs the threshold was 0.3, and such pairs cast no vote at all when their total accumulated support was below 1. Votes were weighted: 1 by default; 20 when the pairing was near-unanimous ($s \leq 0.1$ with total support ≥ 1 , or one pairing accumulated less than 1 unit of support while the other accumulated at least 1); and 0.1 when the voting variant was a tandem-repeat indel. [The weight-20 branch in `PhasingGraph.cpp` was chained off `if(debug)` at `cc17fb1`; the fix is applied in the working tree of the source repository (uncommitted as of 2026-09-20) and must be committed before the release. `findBestEdgePair` still takes an unused `isONT` argument.] The vote was translated into a haplotype vote for v using the already assigned haplotype of u , and the two weighted haplotype totals $h_1(v)$ and $h_2(v)$ were accumulated from all upstream voters. A guard against a single erroneous ultra-long read counted, among the votes received by v , those whose total support was at most 1 read-equivalent; when more than three such votes existed and at least one qualifying vote remained, h_1 and h_2 were recomputed from only those

votes with $s < 0.2$, weight ≥ 1 and a voting variant that is not a small indel. Variant v was assigned to the haplotype with the larger total. When $h_1 = h_2$ (including the case of no votes at all) and v lay beyond the last position reached by any accepted edge, a new phase block was opened at v ; a tie at a position already covered by an accepted edge left the previous block intact and the tied variant unphased, casting no votes. Blocks were identified by the position of their first variant, which defines the PS tag. The six constants of this rule were fixed by a sweep on HG002 at 10–60 $\times$ before the experiments reported here and were not re-tuned afterwards; Supplementary Method 3 states them and Supplementary Fig. 9 shows the sweeps.

Read-based correction

After the initial assignment (Supplementary Fig. 3), every read was assigned to a haplotype by weighted majority over its phased alleles, with SNV and SV alleles contributing 1, indel and tandem-repeat-indel alleles 0.1, and 5mC alleles nothing; a read was tagged only if the majority fraction exceeded 0.65 (`-readConfidence`) and the total weighted evidence exceeded 1. The allele counts of the two read populations were then used to re-derive each variant’s phase: the fraction of tagged reads consistent with the assignment $\rho = \max(n_1, n_2)/(n_1 + n_2)$, where n_1 and n_2 count haplotype-1 reads carrying the reference allele plus haplotype-2 reads carrying the alternate allele and vice versa, had to exceed 0.75 (`-snpConfidence`); otherwise the variant was left unphased. This second pass removes assignments driven by a small number of chimeric or mis-mapped reads.

Phasing entropy and graph export

For each phased variant we recorded the two weighted haplotype totals h_1 and h_2 (after the single-read guard, where that guard fired) and their normalized Shannon entropy

$$\text{PE}(v) = - \sum_{i \in \{1,2\}} p_i \log_2 p_i, \quad p_i = \frac{h_i}{h_1 + h_2}, \quad (4)$$

which ranges from 0 (unanimous votes) to 1 bit (equal support for both haplotypes), with the convention $0 \log_2 0 = 0$. Because PE is set to 0 when $h_1 + h_2 = 0$, a value of exactly 0 denotes either unanimous support or no votes at all; the two cases are distinguished by H1+H2, and the second arises only at the first variant of a phase block, the one position at which a variant without votes is phased (Supplementary Fig. 2f). PE, h_1 and h_2 are written to the output VCF as the INFO fields PE, H1 and H2. With `-dot`, the per-chromosome phasing graph is written in DOT format as `<prefix>.<chrom>.dot`, one directed edge $u^a \rightarrow v^b$ per accepted voting pair with the haplotype weight accumulated at u at the time the vote was cast as label; these files are the input of the GNN module.

Graph neural network for phase-error detection

The graph neural network (GNN) operates on local windows of the phasing graph and assigns each variant a phase-error score between 0 and 1, the classifier output for the event that its phase assignment disagrees locally with the truth set (Supplementary Figs. 4–6); because training used

class subsampling and a focal loss and the score has not been calibrated, it is not a probability of error. We refer to this step, run by the `gnn` sub-command after `phase`, as GNN correction, to distinguish it from the read-based correction inside `phase` (Supplementary Fig. 3). Only records that are heterozygous and phased in the input VCFs enter the module. Variant records from the SNV/indel, SV and 5mC VCFs were merged and sorted by position so that windows span all co-phased variant classes, and a window was opened around every such variant with $PE \geq 0.8$ (`-pe-threshold`), extending 20 variants (`-window`) on each side of the trigger in that merged list. A window therefore contains at most $2 \times 20 + 1 = 41$ variants and 82 allele nodes. Windows with more than 256 nodes are skipped; at the default window size this cap is never reached, and binds only for `-window` greater than 63.

Node and edge features. Each allele node carries 31 features in four groups: phasing evidence (PE, the genotype and allele indices, the trigger and phase-set indicators, the signed offset from the trigger normalized by the largest absolute offset in the window, the two weighted haplotype totals h_1 and h_2 , and six transforms of them); genomic context, computed from the reference when `-r` is given and zero otherwise (homopolymer length, GC content, tandem-repeat unit count, local sequence entropy, variant density and distance to the nearest indel); graph structure (out-degree, phase-set size, log offset from the trigger, dispersion of the outgoing edge weights and a bridge-vertex indicator); and four one-hot variant-class indicators. Each edge carries six features derived from the DOT vote weight w , the genomic distance, the allele-index match and the phase-set relation. Edges are restricted to pairs with both endpoints inside the window, symmetrized, and augmented with one self-loop per node whose features are the mean of its non-self incident edges. Supplementary Tables 1 and 2 give every feature with its exact definition, normalization and cap, and Supplementary Method 4 gives the transforms and the provenance of h_1 and h_2 .

Architecture. The network follows the general, powerful, scalable (GPS) graph-transformer recipe³⁹. Node features were standardized by a batch-normalization layer whose statistics are frozen at inference, projected linearly to a 128-dimensional embedding and passed through a GELU non-linearity. Four identical layers follow, each combining a local branch, a four-head GATv2 message-passing layer over the window’s edges^{40,41}, with a global branch, four-head scaled dot-product self-attention over all nodes of the window⁴². The two branch outputs are added to the residual stream and layer-normalized, then passed through a two-layer feed-forward block (128→256→128, GELU) with a second residual connection and layer normalization. The final node embedding is concatenated with two edge-context scalars and passed through a two-layer classifier (130→128→2) with softmax output.

The model comprises 687,358 float32 values (2,749,432 bytes). Inference is implemented in plain C++ with dense $N \times N$ adjacency and edge tensors and no external runtime dependencies. Predictions for the two allele nodes of a variant were averaged, and a variant covered by several windows received the running mean of its predictions. Only variants in the central zone of a window were scored, namely those no further from the trigger than half the window’s largest trigger distance; this is the same zone the network was trained to score, and nodes outside it are read as context but never scored. The attention equations, the edge-context definition, the requirement for the exact erf-based GELU and the central-zone rule are given in Supplementary

Method 4; the hyperparameters are in Supplementary Table 3.

Training data. Training windows were produced from LongPhase 2 co-phasing runs on the same HG002 ONT R10.4.1 alignments that the evaluation uses, at the six coverages 10, 12, 14, 16, 18 and 20 $\times$ and at all ten down-sampling replicates of each, that is from 60 phasing runs. Each run contributed its SNV/indel, SV and 5mC VCFs together with the per-chromosome DOT export, and windows were opened by the same rules as at inference (`-pe_threshold 0.8`, `-window 20`); the genomic-context features were computed from the same GRCh38 no-alt analysis set. Because every window lies within a single chromosome, the resulting 722,116 windows could be split by chromosome without any window straddling a boundary: chr1–chr14 and chr18–chr20 for training (602,788 windows, 18,356,880 labelled allele nodes in the scored zone), chr15 and chr16 for validation (67,880 windows, 1,509,576 nodes) and chr17, chr21 and chr22 for testing (51,448 windows, 1,367,580 nodes). The training script verifies that the three chromosome sets are disjoint and aborts otherwise. All windows were drawn from the four-class co-phasing configuration, in which SV and 5mC records enter the graphs, but only as context: the truth set used for labelling contains small variants alone, leaving every SV and 5mC node unlabelled, and among scored nodes only the SNV and indel indicators vary. At inference the module makes no such distinction: a variant’s class enters only through the four indicator features, and SV and 5mC records are scored and unphased in the same way as SNVs and indels. Corrections to SV and 5mC records are therefore unsupervised: they rest on transfer from the small variants that share their window, and their extent is reported in the Results (Fig. 4c,d).

Labels. Labels were derived from the v5.0q benchmark restricted to chr1–chr22 small variants, the same truth set used for evaluation, and describe the phase of a variant relative to its neighbours rather than in absolute terms. Within each window the variants were grouped by phase set, the first variant of a group present in the benchmark fixed that group’s orientation, and later variants were labelled by whether their allele-to-haplotype orientation agreed with that anchor. Maximal runs of disagreement were then resolved so that a run is charged to the event that created it, not to each of its members: a single disagreeing variant was labelled *unphase* if its PE was at least 0.9 and *flip* otherwise, and in a run of two or more the first variant, the switch point, was labelled *break* and the remaining members were reset to *correct*. Variants absent from the benchmark, and all SV and 5mC records, were left unlabelled: they were masked from the loss but remained in the graph as context. The released model was trained in binary mode, in which *flip*, *unphase* and *break* are merged into a single *error* class, matching the single action the module is allowed to take. Supplementary Method 4 gives the labelling rules in full.

Class balance and loss. Among the labelled central nodes of the training split, 17,812,710 (97.04%) were *correct* and 544,170 (2.96%) *error*, and 21.5% of windows contained at least one error node. The imbalance was handled by subsampling, not by re-weighting the full distribution: each minibatch kept all of its error nodes and drew correct nodes uniformly at random to at most four times that number (`-ratio_corr 4`), and because the draw was independent at every step the network saw the whole set of correct nodes over training. The cross-entropy on that subset was focal-weighted⁴³ by $(1 - p_t)^\gamma$ with $\gamma = 1.5$ and used label smoothing of 0.05. Supplementary Method 4 explains why the error prevalence here is far above the genome-wide switch-error rate.

Training procedure. Each window graph was made undirected before training by adding, for

every DOT edge, a reverse edge carrying identical features. Read evidence is symmetric, but the DOT export is strictly one-directional (source position below target position), and without the reverse edges a node would receive messages only from upstream. Two label-preserving augmentations, a haplotype swap and a coverage down-scaling, were applied to training windows only (Supplementary Method 4). Optimization used AdamW⁴⁴ with a peak learning rate of 3×10^{-4} and no warm-up, cosine annealing to 10^{-5} over a 100-epoch horizon, gradient-norm clipping at 1.0, and minibatches of 64 windows batched as disjoint graphs; the remaining optimizer settings are the PyTorch defaults and are listed with the rest of the training configuration in Supplementary Table 4. Dropout of 0.4 was applied to the output of the local branch, to the output of the global branch and inside each feed-forward block; dropout is inactive at inference and therefore does not appear in the exported graph. The run was seeded once (`-seed 42`, seeding both the Python and the PyTorch generators) and stopped early after 25 epochs without improvement of the validation criterion, at epoch 69 of the 100-epoch budget. Training used PyTorch 2.11.0 with PyTorch Geometric 2.7.0 under Python 3.12, CUDA 13.0 and cuDNN 9.19, on one NVIDIA GeForce RTX 5090. All reported `gnn` results come from this single training run; sensitivity to reseeding was not assessed.

Model selection. Checkpoints were selected on the validation chromosomes by a net-gain criterion instead of validation loss or F1: at each epoch the predicted phase-error score was thresholded over 81 values spanning $[0.01, 0.99]$ and the score $S = R(2 - 1/P)$ was maximized subject to a recall of at least 0.005, where P and R are the precision and recall of the error class. S is the net fraction of true errors removed when a detection is worth one error and a false detection costs one, and is positive only for $P > 0.5$. The best checkpoint was epoch 44 ($S = 0.0235$, $P = 0.626$, $R = 0.058$, threshold 0.868). On the held-out chromosomes chr17, chr21 and chr22 it attained an accuracy of 0.941 with error-class precision, recall and F1 of 0.258, 0.499 and 0.340 under the argmax decision, over 41,378 labelled error nodes. The threshold applied by `gnn`, 0.30, is far more permissive, because the module’s only action is to withhold phase: a false detection costs one phased variant rather than one switch error. Supplementary Method 4 gives the full selection procedure. [State how 0.30 was chosen: a sweep, on which data and against which objective, or an a-priori choice. Add the phased-fraction/switch-error trade-off across thresholds as a supplementary panel.]

Export and numerical agreement. The trained network was re-implemented with dense tensors, exported to ONNX and embedded in the binary as a base64-encoded float32 blob, and the forward pass refuses to run if the blob length disagrees with the expected parameter count. Numerical agreement was verified at each link of that chain; the largest observed difference in predicted phase-error score was 3×10^{-7} , between the compiled C++ and ONNX Runtime on six synthetic graphs of 3–120 nodes. Supplementary Method 4 describes the export chain and the acceptance bounds.

Hold-out design. The hold-out is at the level of the chromosome. No window from chr17, chr21 or chr22 entered any gradient, any batch-normalization statistic or the checkpoint selection, and chr15 and chr16 were used only for selection. Nothing else was held out. The training windows come from the same 60 down-sampling replicates at 10–20 $\times$ that Figs. 2–4 evaluate, the same v5.0q benchmark supplied both the training labels and the evaluation truth, and the

figures are computed genome-wide over chr1–chr22, of which 17 contributed training windows. No training window was drawn above 20×, although results are reported to 60×, and all training data, and all evaluation data except the PacBio HiFi experiment, come from one individual (HG002) on one platform and chemistry; the HiFi experiment uses the same individual. The consequences for generalization are discussed in the Discussion.

Correction. A variant whose mean predicted error probability was ≥ 0.30 (`-break-threshold`) was unphased: its genotype was rewritten in unordered, ascending form (0|1 and 1|0 both become 0/1) and its PS tag was removed, while the pair of alleles was preserved and no genotype was ever flipped. Bridge vertices may optionally be protected (`-respect-bridge`; off by default). After unphasing, the remaining phased variants of each phase set were checked for connectivity through the surviving DOT edges; when a phase set became disconnected, the largest component kept the original PS and every other component received a new PS equal to the 1-based position of its first variant (`-no-split-blocks` disables this). The corrected SNV/indel, SV and 5mC VCFs are written separately, as `<prefix>.vcf`, `<prefix>_SV.vcf` and `<prefix>_mod.vcf`.

Haplotagging

Reads were assigned to haplotypes by `haplotag`. For each primary alignment with mapping quality ≥ 1 (`-qualityThreshold`), and for supplementary alignments under `-tagSupplementary`, the module counted the phased SNV, indel, SV and 5mC alleles the read carries on each haplotype, every informative observation contributing one vote regardless of class, and tagged the read with `HP:i:1` or `HP:i:2`, the phase set PS and a Phred-scaled phasing quality PQ when the majority haplotype accounted for at least 60% of the informative alleles (`-percentageThreshold`). Two conditions left a read untagged: no clear majority, and informative variants belonging to more than one phase set, for which a majority has no meaning. SV alleles were re-derived from the CIGAR string with the same window and length tolerance as in `phase`, and 5mC evidence was matched by read name from the MR list of the phased 5mC VCF. The evidence model, which weights the four variant classes equally and therefore differs from the read-based correction inside `phase`, and the definition of PQ are given in Supplementary Method 5.

Benchmark data

HG002 (NA24385) Oxford Nanopore R10.4.1 reads were those of the Oxford Nanopore Technologies open-data release of the GIAB Ashkenazi trio sequenced with Ligation Sequencing Kit V14 (`giab_lsk114_2022.12`): two PromethION R10.4.1 flow cells (FLO-PRO114M; PAG65784 and PAG68757) run with SQK-LSK114 at 400 bases per second and 4 kHz sampling in November 2022, basecalled in simplex mode by Oxford Nanopore with Dorado using the `dna_r10.4.1_e8.2_400bps_sup@v4.0.0` model and the `5mCG_5hmCG@v2` modified-base model, with a read N50 of 29.2 and 29.4 kb for the two flow cells (MinKNOW run reports). The reads were aligned to the GRCh38 no-alt analysis set without decoys (`GCA_000001405.15_GRCh38_no_alt_analysis_set`) with minimap2 v2.30 using `-ax map-ont -y`. The full data set (mean autosomal depth 70.4×, from the mosdepth depth distribution released with Oxford Nanopore’s alignment of the same reads to the same reference) was down-

670 sampled with `samtools view -s` (samtools v1.23.1)⁴⁵ to 10, 12, 14, 16, 18, 20, 30, 40, 50 and
671 60×; for 10–20×, ten independent random seeds were drawn and all metrics are reported as the
672 mean over the replicates, whereas 30–60× used a single replicate ([give the ten seed values, or
673 state that they were 1–10, to make the down-sampling reproducible]). Variants were called sepa-
674 rately on every down-sampled alignment, for each coverage and each replicate, and no call set was
675 carried over from another coverage: SNVs and indels with PEPPER-Margin-DeepVariant⁹[give
676 the PEPPER-Margin-DeepVariant release and the model preset (for example `-ont_r10_q20`),
677 and state that its phased output, if enabled, was discarded so that every phaser started from
678 unphased genotypes], SVs with Sniffles2 v2.8.0 with `-output-rnames`, and allele-specific 5mC
679 sites with `longphase modcall`. Phasing accuracy was assessed against two truth sets: the GIAB
680 v4.2.1 small-variant benchmark for HG002²⁹ (HG002_GRCh38_1_22_v4.2.1_benchmark.vcf.gz
681 with HG002_GRCh38_1_22_v4.2.1_benchmark_noinconsistent.bed, from the GIAB FTP
682 release directory `ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/`
683 `NISTv4.2.1/GRCh38/`), and the small-variant benchmark that GIAB derives by aligning the
684 complete diploid T2T-HG002 assembly (Q100 project, T2T-HG002 v1.1)^{30,46,47} to GRCh38,
685 distributed as a draft of the GIAB v5 release³² and referred to throughout this paper as v5.0q.
686 The T2T-HG002-derived benchmark adds 701 Mb of autosomal sequence that v4.2.1 does not
687 assess, most of it satellite and segmentally duplicated³⁰. Identical phased VCFs were scored
688 against both truth sets, each restricted to the regions of its own benchmark BED and to chr1–
689 chr22; chrX, chrY and chrM enter no metric.

690 Comparison with other methods

691 WhatsHap v2.8⁷ and HapCUT2 v1.3.4⁸ were run on the same alignments and the
692 same PEPPER-Margin-DeepVariant calls; for the SNV comparison WhatsHap was run
693 with `-only-snvs`, and HapCUT2 phased SNVs only ([give the complete command line
694 for each, including WhatsHap phase options such as `-ignore-read-groups` and any
695 `-indels/-distrust-genotypes` setting, and the HapCUT2 two-step invocation `extractHAIRS`
696 `-ont 1` followed by `HAPCUT2`, and whether HapCUT2’s default likelihood-based pruning was
697 left on; “default parameters” is not reproducible for either tool.]). LongPhase 1.0¹¹ (release
698 tag v1.0, commit fe8c1fdff11ba2a082d363a7ffcb5c923cacc316) and LongPhase 2 were run
699 with `-ont`; the GNN module was applied to the LongPhase 2 output with default thresholds
700 ($PE \geq 0.8$, break threshold 0.30, no bridge protection, block splitting enabled). For co-phasing
701 experiments, the SNV, SNV+indel, SNV+SV, SNV+5mC and SNV+indel+SV+5mC configu-
702 rations were phased separately, and the GNN was applied to each. Cross-tool comparisons used
703 the SNV-only configuration of LongPhase 2 and were scored on SNVs. Because WhatsHap and
704 HapCUT2 can phase indels but not SVs or 5mC sites, indel, SV and 5mC co-phasing were eval-
705 uated as LongPhase 2 with and without GNN correction, and indel co-phasing was additionally
706 compared with WhatsHap run with its default indel phasing. Margin v2.3.1⁹, Ralphi²⁷ and
707 GCphase³³ phased the same SNV calls at 10–20×, and MethPhaser²¹ was run on the uncor-
708 rected SNV-only output of LongPhase 2 for replicate 1 at every coverage ([versions and command
709 lines for Margin, Ralphi, GCphase and MethPhaser; Supplementary Table 5]). HG002 PacBio
710 HiFi (Revio) data were down-sampled to 10, 20, 30, 40 and 50× with one replicate each and

phased SNV-only with LongPhase 2, with and without correction, and WhatsHap; the network was applied with the weights trained on nanopore data ([HiFi read source and accession, aligner and preset, variant caller and model, LongPhase 2 mode (-pb), and whether one call set was used for every coverage]). [State that all tools ran on the same machine, and whether LongPhase 1.0 was re-run on every down-sampling replicate at 10–20× (the LongPhase 2, WhatsHap and HapCUT2 tables hold every replicate).]

Evaluation metrics

Metrics (Supplementary Fig. 8) were computed with `longphase compare`, a C++ re-implementation of the `whatshap compare` algorithm⁷ that is parallelized over chromosomes with OpenMP. For each chromosome, heterozygous variants common to the truth and query VCFs were intersected by position and alleles and partitioned into blocks that are phased in both files; singleton blocks were ignored. Within each block the query haplotype string was compared with the truth string in switch encoding (the sequence of relative phase changes between consecutive variants), giving the number of *switch errors*; the *switch error rate* is the number of switch errors divided by the number of assessed variant pairs. The *Hamming distance* is the block-wise minimum over the two possible haplotype orientations of the number of misassigned variants, summed over blocks and expressed as a percentage of compared variants. Switch errors were further decomposed into isolated switches and flips (two consecutive switches). The *phased fraction* is the number of truth heterozygous variants phased by the query divided by the total number of truth heterozygous variants, reported separately for SNVs, indels, SVs and 5mC sites; the *block N50* is the span-weighted median block length of the query’s own phase sets of at least two variants, pooled over all chromosomes and independent of the truth set[`compare` applies a `-regions/-regions-bed` filter to the query as well as the truth when it reads them (`CompareProcess.cpp:353`), which would make N50 depend on the benchmark BED. N50 is identical under v4.2.1 and v5.0q in every run, which suggests the BED was applied to the truth VCF beforehand rather than passed to `compare`; state the exact commands]. Switch errors and switch error rates were computed on SNVs only, over pairs of consecutive SNVs within each block, keeping them independent of which additional variant classes were co-phased. The Hamming distance, by contrast, was computed over all heterozygous variants common to truth and query, and therefore includes indels for configurations that phase them; its denominator is the number of compared variants, not the number of assessed SNV pairs[add a Supplementary table defining numerator, denominator, variant classes, truth eligibility and block handling for every metric, and re-score the indel-containing configurations with an SNV-only Hamming distance]. Switch-error positions were exported as BED and used for the analysis of unphased variants in Fig. 6b. Supplementary Method 6 defines each metric precisely. [Define the depth and informativeness bins used in Supplementary Fig. 17b (“low”, “normal”, “high” relative to the genome-wide mean, and the informativeness cut-points) and the “low genotype quality” threshold used in Supplementary Fig. 17c.]

Analysis of unphased SNVs

An SNV was counted as unphased by the correction when it was phased (|) in the `phase` output and unphased (/) in the `gnn` output of the same run. Each such SNV was classified against the v5.0q benchmark VCF as absent, homozygous or heterozygous (Supplementary Fig. 16a). For the 60× run, each was also classified by the T2T-HG002 v1.1 assembly as projected onto GRCh38 by GIAB with dipcall (GRCh38_HG2-T2TQ100-V1.1_dipcall-z2k.dip.bed and .dip.vcf.gz, from the v5.0q release directory): outside the BED, in which both assembled haplotypes align 1:1 to GRCh38 (unassessed); inside it without an assembly variant; or inside it with one. A random sample of 50,000 SNVs that remained phased served as background, and enrichment was summarized per chromosome and in 1-Mb bins. The analysis uses only `awk`, `sort`, `join`, `comm` and `shuf` and is reproduced by `run_t2t_validation.sh`. To compare what LongPhase 2 and WhatsHap leave unphased, heterozygous SNVs present in both output VCFs were partitioned into those unphased by LongPhase 2 only (split by whether `phase` or `gnn` left them unphased), by WhatsHap only and by both, and the fraction of each set inside the other tool’s switch-error intervals was compared with the fraction of all that tool’s assessed SNVs inside those intervals ([define a switch-error interval: the span between the two variants flanking a switch, or a fixed window; and deposit the scripts for Fig. 6 and Supplementary Fig. 16a, which exist only as figure drafts]).

Runtime and memory

Wall-clock time and peak resident memory were measured with `/usr/bin/time -v` on [CPU model and clock, physical core and socket count, RAM, storage type, operating system and kernel, compiler and optimization flags], using `[t]` threads for every tool. [State whether the machine was otherwise idle, how many repeats were timed and whether the reported value is the mean, median or minimum, and report peak resident memory alongside every runtime, since the Results section currently quotes runtimes only; the runtime panel was dropped from the regenerated Fig. 2 for want of an exact source.] GNN correction time was measured separately and added to the LongPhase 2 total. Reported LongPhase 2 times include reading of the alignments, graph construction, phasing and writing of the VCF, and exclude variant calling, alignment and down-sampling.

Statistics and reproducibility

No statistical tests were used to compare methods, and no sample-size calculation, randomization or blinding was applicable: all metrics are deterministic functions of the input data and the only stochastic element is the choice of reads during down-sampling. The replicate design was as follows. At 10, 12, 14, 16, 18 and 20×, ten independent down-sampling replicates were generated from the same full-coverage alignment with ten distinct random seeds, and every reported value at those coverages is the mean over those replicates. Every tool, including LongPhase 2 at 10×, was run on all ten replicates (Supplementary Data 1). At 30, 40, 50 and 60×, and at every HiFi coverage, a single replicate was generated and analysed. No dispersion can be reported at those coverages, and differences between methods there are not accompanied by

a measure of replicate variability. Figs. 2–4 and Supplementary Figs. 10, 13 and 15 draw the replicate s.d. at 10–20×. The down-sampling replicates are technical replicates of one individual, and the tools compared at a given replicate phase the same sampled reads[report fold changes and correction effects as paired, replicate-wise differences with their dispersion, rather than ratios of means]. The output of `phase` does not depend on the thread count by construction: the program parallelizes over chromosomes, each chromosome is processed from start to finish by one thread, and the HTSLib thread pool serves only to decompress the alignments, which are consumed in file order (`PhasingProcess.cpp:96`; `ParsingBam.cpp:1012`). `compare` likewise evaluates every intersection block independently and sums integer counts. The GNN module performs no sampling and uses fixed weights, and its output is deterministic up to floating-point summation order: the phase-error score of a variant covered by several windows is averaged in the order in which threads complete, which can change the last bits of the mean (of order 10^{-7}) and hence the decision only for a variant whose mean lies within that distance of the 0.30 threshold.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The GRCh38 no-alt analysis set used as the reference is distributed by NCBI as `GCA_000001405.15_GRCh38_no_alt_analysis_set.fna.gz` (GenBank assembly GCA_000001405.15), at https://ftp.ncbi.nlm.nih.gov/genomes/all/GCA/000/001/405/GCA_000001405.15_GRCh38/seqs_for_alignment_pipelines.ucsc_ids/. The GIAB HG002 v4.2.1 small-variant benchmark (`HG002_GRCh38_1_22_v4.2.1_benchmark.vcf.gz` and `HG002_GRCh38_1_22_v4.2.1_benchmark_noinconsistent.bed`) is at https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/release/AshkenazimTrio/HG002_NA24385_son/NISTv4.2.1/GRCh38/. The v5.0q benchmark is the GRCh38 small-variant call set of the GIAB draft benchmark release `NIST_HG002_DraftBenchmark_defrabbV0.020-20250117` (`GRCh38_HG2-T2TQ100-V1.1_smvar.vcf.gz`, with the accompanying `GRCh38_HG2-T2TQ100-V1.1_smvar.benchmark.bed`), at https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/data/AshkenazimTrio/analysis/NIST_HG002_DraftBenchmark_defrabbV0.020-20250117/; the file scored here, `HG002_GRCh38_v5.0q_smvar.chr1_22.vcf`, is the chr1–chr22 subset of that VCF and was scored within the accompanying BED. The HG002 nanopore reads are from the Oxford Nanopore open-data release `giab_lsk114_2022.12` (`s3://ont-open-data/giab_lsk114_2022.12/`; raw signal under `flowcells/hg002/`, Dorado `sup@v4.0.0` basecalls with modified-base tags and their alignment under `analysis/wf-human-var-output/hg002_sup_v4/`), distributed under the Creative Commons Attribution-NonCommercial 4.0 International licence (CC BY-NC 4.0). [Still required. (i) State the command used to derive the chr1–chr22 subset of the v5.0q VCF. (ii) Source data for Figs. 2–6 and for every Supplementary Figure with plotted values. Nature Methods requires that restrictions, if any, are stated; if there are none, say so explicitly.]

Code availability

LongPhase 2, including the `modcall`, `phase`, `gnn`, `haplotag` and `compare` sub-commands, is open source under the GNU General Public License v3.0 at <https://github.com/twolinin/longphase>. A Dockerfile, pre-compiled Linux x86-64 binaries and a demonstration data set (chr1:121.5–122.0 Mb) are provided in the repository, together with a script that runs `modcall`, SNV-only and four-class `phase`, `gnn` correction of both and `compare` against a truth VCF in under two minutes on four threads. [Give the exact release tag and the full 40-character commit hash of the version used for every result in this paper. The latest tagged release at the time of writing is v2.0.2 (`main.cpp`:9; `README.md`:34), but v2.0.2 predates the `gnn` and `compare` sub-commands, which exist only on the unmerged JH branch at commit `cc17fb1`. No released version can currently reproduce any `gnn` or `compare` result in this paper. Merge and tag before submission.] The complete training chain is provided as a separate archive, comprising the code for window and feature extraction, training and model selection, ONNX export and weight embedding, together with the driver scripts, the released checkpoint in both PyTorch and ONNX form, the training log and a pinned list of package versions. [These files exist but are not yet in the public repository or in any archive. Deposit them in the repository under an OSI-approved licence and cite them here; Nature Methods will not accept a machine-learning method whose training code is unavailable. Note that the scripts as written carry absolute paths of the development machine (`/ssd/longphase_process/...`, `/disk/software/...`); parameterize them before deposition.] [Deposit `figures-source/make_results_figs.py`, which generates Figs. 2–6 and Supplementary Figs. 10–16 from Supplementary Data 1, together with `run_t2t_validation.sh`; the scripts behind Fig. 6 and Supplementary Figs. 11, 16a and 17 (composition, give-up sets, MethPhaser scoring, depth and entropy classes) are still missing. The schematic Supplementary Figs. 1–8 are generated by `figures/supp/make_supp_figs.py` in the manuscript repository and can be deposited as is.]

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